

Information-Topological Theory: Minimal Foundational Formulation

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Abstract

We present the minimal foundational formulation of Information-Topological Theory (ITT). By deliberately isolating standard mathematical invariants from ITT-specific physical postulates, we establish the topological level selection $k_{\min} = 6$ purely through a $\mathbb{Z}_6$ -equivariant descent condition on the configuration space $L(6, 1) = S^3/\mathbb{Z}_6$, without relying on phenomenological Standard Model parameter fits.

1 Information-Topological Theory: Minimal Foundational Formulation

1.1 Scope and Logical Status

The Information-Topological Theory (ITT) is formulated as a timeless canonical gauge theory on a three-dimensional configuration manifold. The fundamental configuration space is the gauge-orbit space

$$\mathcal{C}_{\text{ITT}} = \mathcal{A}(S^3)/\mathcal{G}(S^3), \quad (1)$$

where $\mathcal{A}(S^3)$ denotes the space of smooth $SU(2)$ connections on S^3 and $\mathcal{G}(S^3)$ the corresponding gauge transformation group. The theory deliberately separates standard mathematical results from ITT-specific postulates. In particular, the integer quantization of the Chern–Simons level is standard; the selection of the minimal level $k = 6$ is an ITT equivariant-descent condition.

1.2 I. Timeless Canonical Configuration Space

The fundamental canonical variables are

$$A_i^a(x), \quad E_a^i(x), \quad (2)$$

with canonical Poisson brackets

$$\{A_i^a(x), E_b^j(y)\} = \delta_i^j \delta_b^a \delta^{(3)}(x - y), \quad (3)$$

and

$$\{A_i^a, A_j^b\} = 0, \quad \{E_a^i, E_b^j\} = 0. \quad (4)$$

No external physical time is postulated at the fundamental level. A parameter τ may be introduced only as a bookkeeping parameter for curves in configuration space.

1.3 II. Chern–Simons Functional and ITT Constraint

Let

$$F = dA + A \wedge A \quad (5)$$

and

$$B_a^i = \frac{1}{2} \epsilon^{ijk} F_{jk,a}. \quad (6)$$

The $SU(2)$ generators are normalized by

$$T_a = \frac{\sigma_a}{2i}, \quad \text{tr}(T_a T_b) = -\frac{1}{2} \delta_{ab}. \quad (7)$$

Define the Chern–Simons functional

$$S_{\text{CS}}^{(0)}[A] = \int_{S^3} \text{tr} \left(A \wedge dA + \frac{2}{3} A \wedge A \wedge A \right), \quad (8)$$

with functional derivative

$$\frac{\delta S_{\text{CS}}^{(0)}}{\delta A_i^a} = -B_a^i. \quad (9)$$

The fundamental ITT constraint is

$$\boxed{\mathcal{C}_a^i \equiv E_a^i + \frac{k}{4\pi} B_a^i = E_a^i - \frac{k}{4\pi} \frac{\delta S_{\text{CS}}^{(0)}}{\delta A_i^a} \approx 0.} \quad (10)$$

The corresponding first-order timeless action is

$$S_{\text{ITT}} = \int d\tau \int_{S^3} d^3x \left[E_a^i \frac{dA_i^a}{d\tau} - N_i^a \mathcal{C}_a^i - A_0^a \mathcal{G}_a \right]. \quad (11)$$

1.4 III. Constraint Algebra

Because $\mathcal{C}$ is a canonical momentum shift by an exact functional one-form,

$$\boxed{\{\mathcal{C}_a^i(x), \mathcal{C}_b^j(y)\} = 0} \quad (12)$$

on the closed manifold S^3 . The magnetic field satisfies the non-Abelian Bianchi identity

$$D_i B_a^i \equiv 0. \quad (13)$$

Consequently,

$$D_i \mathcal{C}_a^i = D_i E_a^i \equiv \mathcal{G}_a. \quad (14)$$

Thus the Gauss constraint is a differential consequence of the ITT constraint:

$$\boxed{\mathcal{G}_a = D_i \mathcal{C}_a^i} \quad (15)$$

At the classical level the constraint system is therefore first-class with a differential reducibility relation. At the quantum level the corresponding identity is understood with a fixed gauge-covariant operator ordering.

1.5 IV. Formal Exact Chern–Simons State

Canonical quantization is defined by

$$\hat{E}_a^i = -i \frac{\delta}{\delta A_i^a}. \quad (16)$$

The formal wave functional

$$\boxed{\Psi_k[A] = \mathcal{N} \exp \left[i \frac{k}{4\pi} S_{\text{CS}}^{(0)}[A] \right]} \quad (17)$$

satisfies

$$\hat{\mathcal{C}}_a^i \Psi_k[A] = 0 \quad (18)$$

because

$$\hat{E}_a^i \Psi_k = -\frac{k}{4\pi} B_a^i \Psi_k. \quad (19)$$

Consequently, for the positive quadratic constraint

$$\hat{\mathcal{H}}_{\text{ITT}} = \frac{1}{2} \hat{\mathcal{C}}_a^i \hat{\mathcal{C}}_i^a, \quad (20)$$

with a specified regularization and operator ordering,

$$\boxed{\hat{\mathcal{H}}_{\text{ITT}} \Psi_k = 0.} \quad (21)$$

This is a formal exact solution of the ITT canonical constraint. A complete Hilbert-space construction, regularization and self-adjoint realization are separate analytical problems.

1.6 V. Unit Topological Sector

For the $SU(2)$ BPST instanton on $D_R^4 \subset \mathbb{R}^4$, with scale ρ , define

$$c = \frac{R^2}{R^2 + \rho^2}. \quad (22)$$

With the above trace convention and orientation,

$$Q(R) = -\frac{1}{8\pi^2} \int_{D_R^4} \text{tr}(F \wedge F) = 3c^2 - 2c^3 = \frac{R^4(R^2 + 3\rho^2)}{(R^2 + \rho^2)^3}. \quad (23)$$

For

$$A|_{S^3} = c g^{-1} dg, \quad (24)$$

the corresponding Chern–Simons transgression gives the same quantity,

$$\frac{1}{2\pi} S_{\text{CS}}(A|_{S^3}) = 3c^2 - 2c^3. \quad (25)$$

In the asymptotic limit $R/\rho \rightarrow \infty$, one obtains the unit topological sector

$$\boxed{Q \rightarrow 1.} \quad (26)$$

The corresponding APS cylinder representation yields the net spectral-flow index

$$\boxed{\text{SF}(\mathcal{D}_{A_c}) = \text{ind} \left(\mathcal{D}_{[0,1] \times S^3}^{\text{APS}} \right) = 1,} \quad (27)$$

with c interpreted solely as a parameter of a family of three-dimensional gauge configurations and not as physical time.

1.7 VI. Standard Level Quantization

Under a large $SU(2)$ gauge transformation with winding number

$$w \in \pi_3(SU(2)) \cong \mathbb{Z}, \quad (28)$$

the normalized Chern–Simons action shifts by an integer multiple of 2π . Therefore gauge invariance of the quantum amplitude requires

$$\boxed{k \in \mathbb{Z}.} \quad (29)$$

This is a standard Chern–Simons quantization condition and does not by itself select $k = 6$.

1.8 VII. Lens-Space Topological Sector

The ITT quotient geometry is defined by the free diagonal action

$$(z_1, z_2) \mapsto \left(e^{2\pi i/6} z_1, e^{2\pi i/6} z_2 \right) \quad (30)$$

on $S^3 \subset \mathbb{C}^2$. The resulting lens space is

$$\boxed{L(6, 1) = S^3/\mathbb{Z}_6,} \quad (31)$$

with $\pi_1(L(6, 1)) \cong \mathbb{Z}_6$. Let $\rho_1 : \mathbb{Z}_6 \rightarrow SU(2)$ be the fundamental non-trivial flat representation,

$$\rho_1(\gamma) = \begin{pmatrix} e^{2\pi i/6} & 0 \\ 0 & e^{-2\pi i/6} \end{pmatrix}. \quad (32)$$

With the fixed orientation and trace convention, its Chern–Simons invariant is

$$\boxed{\text{cs}(\rho_1) = -\frac{1}{6} \pmod{\mathbb{Z}}.} \quad (33)$$

The value follows from the rational intersection/linking form associated with the (-6) -framed surgery description of $L(6, 1)$.

1.9 VIII. ITT Equivariant Descent Condition

The ITT quantum state is required to admit a $\mathbb{Z}_6$ -equivariant lift to its prequantum line bundle. Denote the generator of $\mathbb{Z}_6$ by ζ .

The ITT equivariant structure is required to satisfy

$$\boxed{\chi_k(\zeta) = \exp(2\pi i k \text{cs}(\rho_1))}. \quad (34)$$

Using $\text{cs}(\rho_1) = -\frac{1}{6}$, the fiber character becomes

$$\boxed{\chi_k(\zeta) = e^{-2\pi i k/6}}. \quad (35)$$

This equation is the defining ITT equivariant-lift condition. It is not asserted as a universal theorem of ordinary $SU(2)$ Chern–Simons theory.

1.10 IX. ITT Level Selection Rule

The ITT physical state is required to descend to the quotient configuration space with trivial stabilizer action. Therefore

$$\chi_k(\zeta) = 1. \quad (36)$$

Hence

$$e^{-2\pi i k/6} = 1 \quad (37)$$

and therefore

$$\boxed{k \in 6\mathbb{Z}}. \quad (38)$$

The minimal positive ITT level is

$$\boxed{k_{\min} = 6}. \quad (39)$$

Thus the level selection is not derived from a phenomenological relation such as 3×2 . It is an ITT-specific equivariant descent condition associated with the chosen $\mathbb{Z}_6$ quotient geometry.

1.11 X. Independent Soliton Sector

The number of particle generations is not used in the derivation of k . Instead, ITT treats

$$N_{\text{gen}} = 3 \quad (40)$$

as an independent dynamical prediction of the soliton sector. The future objective is to construct an ITT energy functional

$$E_{\text{ITT}}[n, A] \quad (41)$$

whose stable topological sectors satisfy

$$Q_H = 1, 2, 3 \quad (42)$$

while higher-charge configurations are dynamically unstable or energetically decomposable. Until such a result is established, $N_{\text{gen}} = 3$ remains an ITT postulate or target prediction and is not used in the proof of $k_{\min} = 6$.

1.12 XI. Foundational Logical Chain

The minimal ITT foundation is summarized by the sequence:

$$\begin{aligned}
& \mathcal{C}_{\text{ITT}} = \mathcal{A}(S^3)/\mathcal{G}(S^3) \\
& \Downarrow \\
& \boxed{\mathcal{C}_a^i = E_a^i + \frac{k}{4\pi} B_a^i, \quad \{\mathcal{C}, \mathcal{C}\} = 0} \\
& \Downarrow \\
& \boxed{D_i \mathcal{C}_a^i = \mathcal{G}_a} \\
& \Downarrow \\
& \boxed{\Psi_k[A] = \mathcal{N} e^{ik S_{\text{CS}}^{(0)}[A]/4\pi}} \\
& \Downarrow \\
& \boxed{L(6, 1) = S^3/\mathbb{Z}_6, \quad \text{cs}(\rho_1) = -\frac{1}{6}} \\
& \Downarrow \\
& \boxed{\chi_k(\zeta) = e^{-2\pi i k/6}} \\
& \Downarrow \\
& \boxed{\chi_k(\zeta) = 1 \implies k \in 6\mathbb{Z}} \\
& \Downarrow \\
& \boxed{k_{\min} = 6.}
\end{aligned} \tag{43}$$

The independent soliton problem begins after this foundational chain:

$$E_{\text{ITT}} \longrightarrow \text{stable Hopf sectors } (Q_H = 1, 2, 3) \longrightarrow N_{\text{gen}} = 3. \tag{44}$$

1.13 Foundational Status

The minimal theory therefore distinguishes three levels of statement:

1. **Standard mathematics:** gauge theory on S^3 , BPST topology, APS spectral flow, Chern–Simons integer level quantization, lens-space topology, and rational Chern–Simons invariants.
2. **ITT defining structure:** the helicity constraint, the timeless canonical formulation, the $\mathbb{Z}_6$ -equivariant quantum structure, and the requirement of trivial equivariant descent.
3. **Future ITT predictions:** the dynamical derivation of $N_{\text{gen}} = 3$, the particle mass spectrum, coupling constants, and independent experimental observables.

The foundational claim of ITT is therefore:

$$\boxed{\text{timeless gauge configuration space} + \text{topological constraint} + \mathbb{Z}_6 \text{ equivariant descent} \implies k_{\min} = 6.} \tag{45}$$

No Standard-Model parameter fit and no phenomenological factor 3×2 is used in this foundational selection of the Chern–Simons level.